

Solving Models with Uncertainty

November 8, 2021

1 Perturbation

In this lab we're going to discuss perturbation methods. Perturbation methods are used in Dynare, a software that is commonly used in the profession. We're first going to discuss the theory behind the method. Then, we will discuss how to write Dynare code. The benefit of Dynare is that you can write your equations just as you would on a piece of paper.

1.1 Theory

Here I am following Judd and Schmitt-Grohe and Uribe. Perturbation methods rely on two fundamental ideas that you have already encountered: Taylor series expansions and the implicit function theorem.

Perturbation works as follows. Suppose we have a function, F such that

$$F(x, \sigma) = 0 \quad (1)$$

where σ is a parameter. Given that σ is a parameter, we can view the x that sets F to 0 for a given σ as a function of σ . That is, we can rewrite the above as

$$F(x(\sigma), \sigma) = 0 \quad (2)$$

Now suppose we want to approximate $x(\cdot)$ around the point $\sigma = 0$. To do so, we can use a first order Taylor approximation

$$x(\sigma) \approx x(0) + x_\sigma(0) \sigma \quad (3)$$

There's only one issue. We don't know x_σ . This is where equation (2) comes in. Using the implicit function theorem, we have

$$\begin{aligned} F_x(x(0), 0) x_\sigma(0) + F_\sigma(x(0), 0) &= 0 \rightarrow \\ x_\sigma(0) &= -\frac{F_\sigma(x(0), 0)}{F_x(x(0), 0)} \end{aligned} \quad (4)$$

Therefore, if we combine this with (3) we have

$$x(\sigma) \approx x(0) - \frac{F_\sigma(x(0), 0)}{F_x(x(0), 0)} \sigma \quad (6)$$

F is a known function and so we will be able to compute F_σ and F_x . Therefore, we have an approximation for $x(\sigma)$. Now, in the case of our problems, $x(0)$ will usually be the steady state, which can we easily solve for either analytically or numerically.

1.2 Model with uncertainty

Now we will discuss how this approach is useful in solving models with uncertainty. In Schmitt-Grohe and Uribe, they use x to denote state variables and y to denote control variables. With this notation, we can write the problem we're trying to solve as

$$E[f(y', y, x', x)] = 0 \quad (7)$$

As you know, what we're looking for in a solution is a set of functions such that

$$y = g(x, \sigma) \quad (8)$$

$$x' = h(x, \sigma) + \eta\sigma\epsilon' \quad (9)$$

We can use the same approach as we did previously. Specifically, we're going to use a first order Taylor series expansion around the point $(\bar{x}, 0)$.

$$g(x, \sigma) \approx g(\bar{x}, 0) + g_x(\bar{x}, 0)(x - \bar{x}) + g_\sigma(\bar{x}, 0)(\sigma) \quad (10)$$

$$h(x, \sigma) \approx h(\bar{x}, 0) + h_x(\bar{x}, 0)(x - \bar{x}) + h_\sigma(\bar{x}, 0)(\sigma) \quad (11)$$

Just as before, we can use the implicit function theroem to find the derivatives. We'll start by finding the partials with respect to σ and leave the others until the next section. Note the following

$$y' = g([h(x, \sigma) + \eta\sigma\epsilon'], \sigma) \quad (12)$$

With this representation of y' we can now find g_σ and h_σ .

$$F_\sigma = E\{f_{y'}[g_x(h_\sigma + \eta\epsilon') + g_\sigma] + f_y g_\sigma + f_{x'}[h_\sigma + \eta\epsilon']\} \quad (13)$$

$$= f_{y'}[g_x h_\sigma + g_\sigma] + f_y g_\sigma + f_{x'} h_\sigma \quad (14)$$

$$= \begin{bmatrix} f_{y'} g_x + f_{x'} & f_{y'} + f_y \end{bmatrix} \begin{bmatrix} h_\sigma \\ g_\sigma \end{bmatrix} = 0 \quad (15)$$

What this last part says is that, if we have a unique solution to our problem (which we are assuming), then h_σ and g_σ must both be 0. Therefore, we have the following

$$y \approx g(\bar{x}, 0) + g_x(\bar{x}, 0)(x - \bar{x}) \quad (16)$$

$$x' \approx h(\bar{x}, 0) + h_x(\bar{x}, 0)(x - \bar{x}) + \eta\sigma\epsilon' \quad (17)$$

Now, given that $E[\epsilon'] = 0$ we see that, using first order perturbation, both y and x are equal to their non-stochastic steady state. That is

$$y = g(\bar{x}, 0) \quad (18)$$

$$x = h(\bar{x}, 0) \quad (19)$$

What this means is that, up to a first order approximation the policy functions do not depend on the variance co-variance matrix of the shocks. As discussed in SGU, first order perturbation cannot be used to discuss things such as risk premium.

1.3 Growth Model with Labor

Lets consider the model with log utility over consumption and labor. We can write our function F as

$$F(y', y, x', x) = \begin{bmatrix} y_1^{-\gamma} - \beta E \left[y_1'^{-\gamma} \left(\alpha e^{x_2'} x_1'^{(\alpha-1)} y_2'^{(1-\alpha)} + (1-\delta) \right) \right] \\ y_1 + x_1' - e^{x_2} x_1^\alpha y_2^{1-\alpha} - (1-\delta) x_1 \\ x_2' - \rho x_2 - \sigma \eta \epsilon' \\ \psi y_2^\varphi - (1-\alpha) e^{x_2} x_1^\alpha y_2^{-\alpha} y_1^{-\gamma} \end{bmatrix} \quad (20)$$

This gives us the following matrices

$$f_{x'} = \begin{bmatrix} -\beta E \left[y_1'^{-\gamma} \left(\alpha e^{x_2'} x_1'^{(\alpha-1)} y_2'^{(1-\alpha)} \right) \right] & -\beta E \left[y_1'^{-\gamma} \left(\alpha (\alpha-1) e^{x_2'} x_1'^{(\alpha-2)} y_2'^{(1-\alpha)} \right) \right] \\ 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$f_{y'} = \begin{bmatrix} \gamma \beta E \left[y_1'^{-\gamma-1} \left(\alpha e^{x_2'} x_1'^{(\alpha-1)} y_2'^{(1-\alpha)} - (1-\delta) \right) \right] & -\beta E \left[y_1'^{-\gamma} \left(\alpha (1-\alpha) e^{x_2'} x_1'^{(\alpha-1)} y_2'^{(-\alpha)} \right) \right] \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$f_x = \begin{bmatrix} 0 & 0 \\ -e^{x_2} x_1^\alpha y_2^{1-\alpha} & -\alpha e^{x_2} x_1^{\alpha-1} y_2^{1-\alpha} - (1-\delta) \\ -\rho & 0 \\ -(1-\alpha) e^{x_2} x_1^\alpha y_2^{-\alpha} y_1^{-\gamma} & -(1-\alpha) \alpha e^{x_2} x_1^{\alpha-1} y_2^{-\alpha} y_1^{-\gamma} \end{bmatrix}$$

$$f_y = \begin{bmatrix} -\gamma y_1^{-\gamma-1} & 0 \\ 1 & -(1-\alpha) e^{x_2} x_1^\alpha y_2^{-\alpha} \\ 0 & 0 \\ \gamma (1-\alpha) e^{x_2} x_1^\alpha y_2^{-\alpha} y_1^{-\gamma-1} & \varphi \psi y_2^{\varphi-1} + \alpha (1-\alpha) \alpha e^{x_2} x_1^{\alpha-1} y_2^{-\alpha-1} y_1^{-\gamma} \end{bmatrix}$$

Now, we have the following system of equations that we need to solve

$$\begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix} \begin{bmatrix} I \\ g_x \end{bmatrix} h_x = - \begin{bmatrix} f_x & f_y \end{bmatrix} \begin{bmatrix} I \\ g_x \end{bmatrix}$$

Let $A = \begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix}$ and $B = - \begin{bmatrix} f_x & f_y \end{bmatrix}$. Next, we write the eigenvalue decomposition of h_x as

$$h_x P = P \Lambda \quad (21)$$

Next we'll define a new matrix, Z as

$$Z = \begin{bmatrix} I \\ g_x \end{bmatrix} P \quad (22)$$

Therefore, we can write the system as

$$AZ\Lambda = BZ \quad (23)$$

We can solve this problem using standard eigenvalue decomposition. Once we have the resulting matrix of eigenvalues D and matrix of eigenvectors V we

proceed as follows. First, sort the eigenvalues from smallest to largest and denote D_{11} as the matrix where all eigenvalues are less than one in absolute value. Note that, for us to have a unique solution to this problem, we need the number of eigenvalues less than one in absolute value to be equal to the number of state variables. Next, sort the eigenvectors in V so that the eigenvectors are in the same column ordering as the eigenvalues. Denote the matrix V_1 as the matrix of eigenvectors corresponding to those that are less than one in absolute value. Now partition V_1 so that the first n_x rows are in V_{11} and the remaining are in V_{12} . Therefore, we have the following

$$\Lambda = D_{11} \quad (24)$$

$$\begin{bmatrix} I \\ g_x \end{bmatrix} P = V_1 = \begin{bmatrix} V_{11} \\ V_{12} \end{bmatrix} \quad (25)$$

Using this in the eigenvalue decomposition of h_x we arrive at

$$h_x = V_{11} D_{11} V_{11}^{-1} \quad (26)$$

Finally, we can solve for g_x and obtain

$$g_x = V_{12} V_{11}^{-1} \quad (27)$$

1.4 Follow Up Questions

- What happens if we have more eigenvalues greater than 1 in absolute value than we have control variables? Do we have a unique solution, indeterminate solution or no solution?
- What happens if we have fewer eigenvalues greater than 1 in absolute value than we have control variables? Do we have a unique solution, indeterminate solution or no solution?